Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_34

1. Summary of Key Deviations

The analysis of Run_34 reveals a complex and severe dual-fault scenario, involving both a major physical blockage and significant over-performance of the pump.

- **Flow_Bypass (Critical):** The most extreme symptom is the bypass flow, which reads up to **+82.1%** higher than healthy. This indicates an enormous amount of fuel is being diverted from the engines.
- **Flow_Pump (Moderate):** The pump flow shows a very large positive **offset**, reading over **+21%** higher than the healthy baseline. This indicates the pump is not only compensating for a blockage but is actively and significantly over-performing.
- **Flow_Engine1-4 (Moderate):** All four engine flows show a major, consistent **negative offset** of approximately **-25%**. This confirms that fuel is being severely restricted from reaching the combustion chamber.
- **Pressure Signals (Moderate):** System pressures (`p_FMU`, `p_Shut`, `p_Combustion`) show a moderate **negative offset**, which is counter-intuitive given the blockage but likely caused by the extreme flow dynamics in the bypass loop.
- **Flow_PRV (Nominal):** There is no significant deviation on the `Flow_PRV` signal.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Fuel Metering Unit (FMU) & Main Fuel Pump**	**Dual Fault: Severe Blockage & Pump Over-performance**	This is the only hypothesis that fits all the evidence. A **severe blockage at the FMU** explains why all `Flow_Engine` readings are low (-25%) and why `Flow_Bypass` is critically high (+82%). Simultaneously, a separate fault is causing the **Main Fuel Pump to massively over-perform** (+21%). The combination of these two faults creates the observed scenario: the over-performing pump generates huge amounts of excess flow, and with the path to the engines blocked, almost all of this excess flow is forced through the bypass loop.

2	**Fuel Metering Unit (FMU)**	**Severe Partial Blockage**	A blockage at the FMU alone would explain the low `Flow_Engine` and high `Flow_Bypass`. However, it does not explain why `Flow_Pump` is reading +21% higher than normal. In a simple blockage case, the pump flow should be closer to nominal.
3	**Main Fuel Pump**	**Severe Over-performance**	Pump over-performance alone would cause both `Flow_Pump` and `Flow_Bypass` to be very high. However, it does not explain why all `Flow_Engine` signals are uniformly low, which proves there must also be a significant downstream blockage.
4	**Bypass Valve**	**Stuck Open**	A stuck-open bypass valve would cause high bypass flow but would lead to a *drop* in pump flow and system pressure, which contradicts the observed +21% pump flow.
5	**Multiple Sensor Failures**	**Various Errors**	It is extremely unlikely that the `Flow_Engine` sensors would all fail with a negative offset while the `Flow_Pump` and `Flow_Bypass` sensors fail with a large positive offset. The pattern is clearly indicative of a complex but physically consistent dual-fault event.

3. Most Probable Hypothesis

The most probable root cause is a **Dual Fault Scenario: (1) a Severe Blockage at the Fuel Metering Unit (FMU) AND (2) a Significant Over-performance of the Main Fuel Pump.**

Reasoning: No single fault can explain the full set of symptoms. The data presents two distinct and overlapping problems. The first is a classic downstream blockage, proven by the low engine flows and extremely high bypass flow. The second is a pump that is delivering far too much fuel (+21%). A simple blockage wouldn't cause the pump to over-perform this dramatically, and simple over-performance wouldn't cause all engine flows to drop. It is the combination of these two independent failures that creates this unique and severe diagnostic signature, where a wildly over-performing pump is forcing its excess flow into a bypass loop because its primary path to the engines is obstructed.